Ex.No:13 NMAP to Discover Live Hosts using ARP scan, ICMP Scan, and TCP/UDP ping scan using TryHackme Platform

Aim:
To discover live hosts using Nmap scans (ARP, ICMP, TCP/UDP) on the TryHackme platform Room link: https://tryhackme.com/room/nmapzol.

Procedure:
Task 1: Introduction
ARP scan: This scan uses ARP requests to discover live hosts.
ICMP scan: This scan uses ICMP requests to identify live hosts.
TCP/UDP ping scan: This scan sends packets to TCP ports and UDP ports to determine live hosts.

Task 2: Subnetworks
Network segment - It is a collection of computers linked together through a common "medium".
Subnetwork - It is usually the equivalent of one or more networks segments linked together and configured to utilize the same router in an IP network.
* subnets with /16
* subnets with /24

Task 3: Enumerating Targets
We need to identify the targets will want to scan before we conduct any "action".

1) list
- <MACHINE-IP> <scanme.nmap.org>
- will scan 3 IP address.

2) Range
- 10.11.12.15.20
- will scan 6 IP address.

3) subnet
- <machine-IP>/30
- will scan 4 IP addresses
nmap -sL <TARGETS>

Task 4: Discovering live host
ARP, from link layer - (commonly used for man-in-middle attack (ARP spoofing and ARP poisoning)
ICMP from network layer - commonly used in DDOS attacks.
TCP from transport layer - SYN flooding, TCP rest, TCP session hijacking attacks.
UDP from transport layer - Also used in DDOS

Task 5: Nmap Host discovery using ARP
When a privileged user tries to scan <A1, NMAP uses all requests.
To discover internet hosts without port scanning live systems use nmap -sn Targets.
nmap -PR -sn <TARGETS>
-PR specifies that you only want an ARP scan.
* sn nmap -PR -sn <MACHINE IP>/24
* sn nmap -PR -sn <ATTACKBOX IP>/24

Task 6: Nmap host discovery using ICMP.
* Add -PE option to utilize ICMP echo requests to discover live hosts.
* nmap -PE -sn TARGET to check host is live.
-PP option instructs Nmap to use ICMP timestamp requests.
nmap -PP -sn TARGET
-P n option can enable this scan to

Task 7: Nmap host discovery using TCP and UDP

Nmap to use TCP SYN ping use -PS followed by the port no, range, list or a combination of this options.

-PS21, -PS23 targets ports 21, 22, 23, 23

nmap -PS-SN TARGET check the TCP port is open.

Task 8: using reverse-DNS lookup

Nmap is to use reverse-DNS in the that -R option to query the DNS server even for offline hosts.

Task 9: summary.

ARP scan, ICMP Echo scan, ICMP time scans, ICMP address, mask scan, TCP SYN ping scan, TCP ACK ping scan, UDP ping scan.

Result:

NMAP to discover live host using the above platform was successfully executed.